

TCS NQT Style Practice Questions

Q1. Shift Negative Numbers to the Left

Problem Statement: A warehouse stores products represented by an array of integers. Negative values represent defective products. Move all negative numbers to the beginning of the array while preserving the relative order of the remaining elements.

Input Format:

First line: N (size of array)

Next N lines: array elements.

Output Format:

Print the rearranged array.

Example:

Input: N=8, [5,-2,7,-1,8,-4,9,3]

Output: -2 -1 -4 5 7 8 9 3

Constraints:

$1 \leq N \leq 100$, $-1000 \leq \text{Arr}[i] \leq 1000$

Q2. Reverse Digits and Find Difference

Problem Statement: Given a positive integer N, reverse its digits and calculate the absolute difference between the original number and the reversed number.

Input Format:

Single integer N.

Output Format:

Print the absolute difference.

Example:

Input: 1234

Reverse: 4321

Output: 3087

Constraints:

$1 \leq N \leq 100000$

Q3. Count Fridays in N Days

Problem Statement: Given the starting day of a month and the number of days in the month, count how many Fridays occur during that period.

Input Format:

First line: starting day (mon/tue/wed/thu/fri/sat/sun)

Second line: number of days N.

Output Format:

Print the number of Fridays.

Example:

Input: wed, 15

Output: 2

Constraints:

$1 \leq N \leq 31$

Q4. Sort Employee Ratings

Problem Statement: Employee ratings are represented using only 1, 2 and 3. Sort the ratings in ascending order.

Input Format:

First line: N

Next N lines: ratings values.

Output Format:

Print sorted ratings.

Example:

Input: [2,3,1,2,1,3,2]

Output: 1 1 2 2 2 3 3

Constraints:

$1 \leq N \leq 100$ and $\text{Arr}[i] \in \{1,2,3\}$

Q5. Count Record-Breaking Temperatures

Problem Statement: Count the number of temperatures that are greater than all previous temperatures. The first temperature is always counted.

Input Format:

First line: N

Next N lines: temperatures.

Output Format:

Print the count of record-breaking temperatures.

Example:

Input: [30,28,32,31,35,34]

Output: 3

Constraints:

$1 \leq N \leq 100$ and $0 \leq \text{Arr}[i] \leq 60$